

3 Sucrose is a disaccharide made from glucose and fructose.

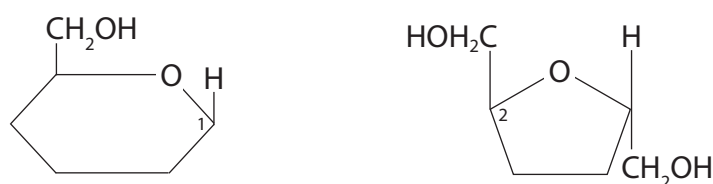
Glucose is joined to fructose by a 1–2 glycosidic bond.

(a) (i) The diagram shows a molecule of glucose and a molecule of fructose.

Carbon 1 on the glucose and carbon 2 on the fructose are numbered.

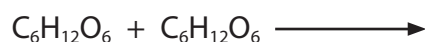
Complete the diagram to show a glycosidic bond between the glucose molecule and the fructose molecule.

(1)



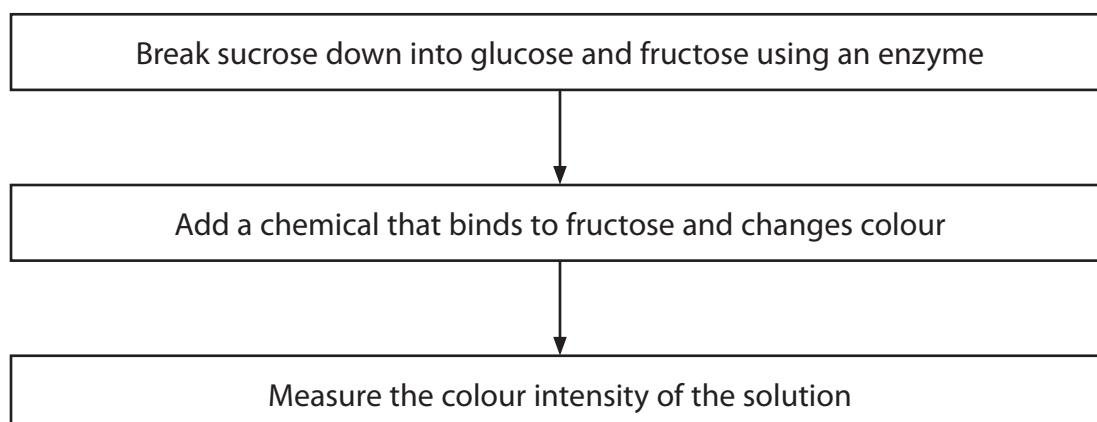
(ii) Complete the formula equation for the reaction that makes sucrose from glucose and fructose.

(2)



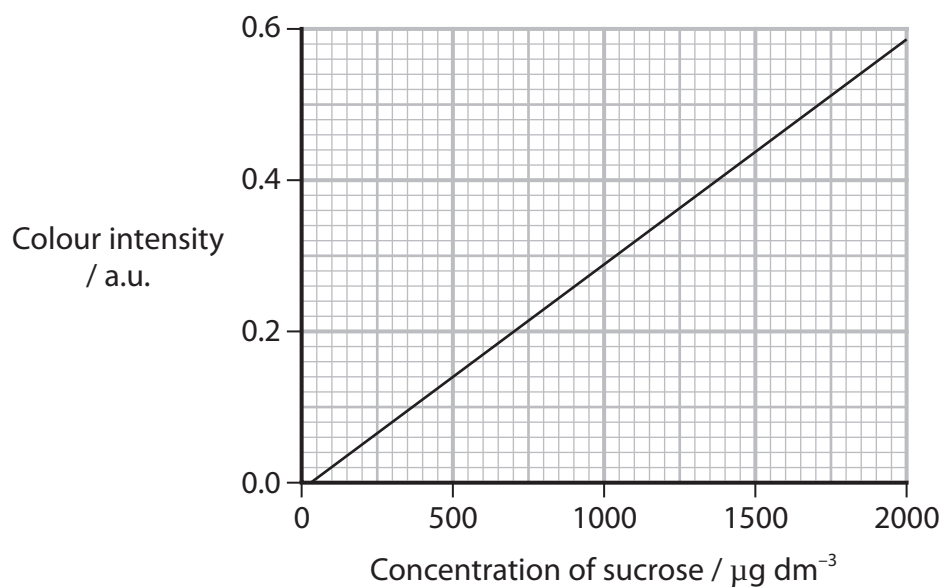
(b) The concentration of sucrose in a solution can be determined by a test.

The chart shows the steps involved in this test.



The colour intensity is directly proportional to the concentration of sucrose.

The graph shows this relationship.



- (i) Give a reason why the colour intensity is directly proportional to the concentration of sucrose.

(1)

- (ii) Suggest why the line does not start at the origin of the graph.

(1)

- (iii) State why the concentration of maltose and lactose cannot be measured using this test.

(1)

(Total for Question 3 = 6 marks)



P 7 1 9 3 4 A 0 7 2 8